

HSUAN-TING LAI

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EDUCATION

Ph.D. student <i>Geophysics</i> Graduate Institute of Geophysics, National Central University	Sep. 2024 – Present Taoyuan, Taiwan
Master of Science <i>Astronomy</i> Graduate Institute of Astronomy, National Central University	Aug. 2019 – July 2021 Taoyuan, Taiwan
Bachelor of Science <i>Physics</i> Department of Applied Physics and Chemistry, University of Taipei	Sep. 2015 – July 2019 Taipei, Taiwan

PROJECTS AND RESEARCH

Work Research Advisor: Prof. Wang, Shih-Jung Geological 3D modeling Project link	Feb. 2022 – Mar. 2024 National Central University
Graduate Research Advisor: Prof. Ip, Wing-Huen - Dynamical evolution and lifetime of Near-Earth Objects (NEOs) under the Yarkovsky effect Thesis link - An Analysis of Regional H_2O and CH_3OH Production rates of Comet 67P from the MIRO Measurements	Sep. 2019 – July 2021 National Central University
Summer Student program & Undergraduate Research Advisor: Prof. Tseng, Wei-Ling Chemical Composition of the Inner Coma of Comet 67P – Rosetta Mission	July 2018 – July 2019 National Taiwan Normal University
Undergraduate Research Advisor: Dr. Lin, Yen-Ting The ratio of multiple cores of brightest cluster galaxies from MaNGA data in redshift 0–0.2	Sep. 2017 – Dec. 2018 Academia Sinica Institute of Astronomy and Astrophysics

PUBLICATIONS

- Lai, H. T.*** & Ip, W. H. (2022). The orbital evolution of Atira asteroids. *Monthly Notices of the Royal Astronomical Society*, 517(4), 5921–5929. [Link](#)
- Hsu, Y. H.*, Lin, Y. T., Huang, S., Nelson, D., Rodriguez-Gomez, V., **Lai, H. T.**, ... & Schneider, D. P. (2022). SDSS-IV MaNGA: Cannibalism Caught in the Act—On the Frequency of Occurrence of Multiple Cores in Brightest Cluster Galaxies. *The Astrophysical Journal*, 933(1), 61. [Link](#)

CONFERENCE PAPERS

- Lai, H.-T.*** and Ip, W.-H. (2022). DYNAMICAL LIFETIME AND EVOLUTION OF NEAR-EARTH OBJECTS UNDER THE YARKOVSKY EFFECT. In *18th Annual Meeting of the Asia Oceania Geosciences Society: Proceedings of the 18th Annual Meeting of the Asia Oceania Geosciences Society (AOGS 2021)*, pp. 183–185. [Link](#)

CONFERENCES AND PRESENTATIONS

- Lai, H.-T.*** and Ip, W.-H. (2026). *Dynamical Migration Pathways of Themis Family Asteroids*. Poster presented at: 15th Asteroids, Comets, Meteors Conference (ACM 2026), July 06–10, 2026, Center for Modern Didactics Poznan University of Technology, Poznań, Poland [Link](#)

Lai, H.-T.*, Ma, K.-F. and Lin, Y.-Y. (2026). *Estimating the Thermal Diffusivity Structure Beneath Borehole A of the Milun Fault Using Long-Term DTS Observations*. Oral presented at: 2026 Joint Annual Meeting and Scientific Conference of the Geophysical Society of the Republic of China and the Geological Society of the Republic of China (Geosciences 2026), June 15–16, 2026, National Chung Cheng University, Chiayi, Taiwan. [Link](#)

Lai, H.-T.*, Ma, K.-F. and Lin, Y.-Y. (2026). *Thermal Characteristics of the Milun Fault Using Borehole Optical Fiber Sensing*. Poster presented at: The Seismological Society of America 2026 Annual Meeting (SSA 2026), April 14–18, 2026, Pasadena Convention Center, Pasadena, California, United States. [Link](#)

Lai, H.-T.* and Ip, W.-H. (2026). *Migration Pathways from the Themis Asteroid Family to the Inner Solar Region*. Poster presented at: Annual Meeting of the Physical Society of Taiwan (TPS 2026), January 13–15, 2026, National Chung Cheng University, Chiayi, Taiwan. [Link](#) [Poster](#)

Ma, K.-F.* and **Lai, H.-T.** (2025). *Characterizing Fault Zone Architecture with Borehole Fiber-Optic Acoustic and Thermal Measurements*. Oral presented at: American Geophysical Union Fall Meeting (AGU 2025), December 15–19, 2025, New Orleans, Louisiana, United States. [Link](#)

Lai, H.-T.*, Ma, K.-F. and Lin, Y.-Y. (2025). *Persistent Thermal Anomalies beneath the Milun Fault Using Borehole Optical Fiber Sensing*. Poster presented at: 12th ACES (APEC Cooperation for Earthquake Science) International Workshop, November 04–06, 2025, Environmental Change Research Building, Academia Sinica, Taipei, Taiwan. [Link](#)

Lai, H.-T.*, Ma, K.-F. and Lin, Y.-Y. (2025). *Thermal Characteristics of the Milun Fault Using Borehole Optical Fiber Sensing*. Oral presented at: The 2025 Taiwan Geosciences Assembly (TGA 2025), June 16–19, 2025, Nangang International Exhibition Center, Hall 1, Taipei, Taiwan. [Link](#)

Lai, H.-T.*, Ma, K.-F. and Lin, Y.-Y. (2025). *Thermal Characteristics of the Milun Fault Using Borehole Optical Fiber Sensing*. Poster presented at: Japan Geoscience Union Meeting 2025 (JpGU 2025), May 24–30, 2025, Makuhari Messe, Chiba, Japan. [Link](#)

Lai, H.-T.* and Ip, W.-H. (2021). *Dynamical evolution and lifetime of Near-Earth Objects (NEOs) under the Yarkovsky effect*. Invited oral presented (Online) at: The 18th Annual Meeting of the Asia Oceania Geosciences Society (AOGS 2021), August 1–6, 2021. [Link](#)

Lai, H.-T.* and Ip, W.-H. (2020). *A comparative study of the orbital evolution of Atira and Vatira asteroids*. Poster presented at: The Astronomical Society of the Republic of China (ASROC), September 3–4, 2020, Institute of Astronomy and Astrophysics, Academia Sinica (ASIAA), Taipei, Taiwan. [Link](#)

Lai, H.-T.*, Ip, W.-H., Tseng, W.-L., Lai, I.-L. and Marshall, D. W. (2020). *An Analysis of Regional H₂O and CH₃OH Production rates of Comet 67P from the MIRO Measurements*. Poster presented at: Annual Meeting of the Physical Society of Taiwan, February 5–7, 2020, National Pingtung University, Pingtung, Taiwan. [Link](#)

Lai, H.-T.*, Tseng, W.-L., Hung, N.-F., Lai, I.-L., and Ip, W.-H. (2019). *A study of the coma composition of the comet 67P*. Poster presented at: Annual Meeting of the Physical Society of Taiwan, January 23–25, 2019; National Chiao Tung University, Hsinchu, Taiwan. [Link](#)

HONORS AND AWARDS

NSTC Graduate Research Fellowship	Oct. 2024
Poster Honorable mention at Annual Meeting of the Physical Society of Taiwan	Jan. 2019
The University Consortium of ALMA–Taiwan Summer Student Program Fellowships	July 2018